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Media Makes Momentum

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Relying on 2.2 million articles from forty-five national and local U.S. newspapers between 1989 and 2010, we find that firms particularly covered by the media exhibit, ceteris paribus, significantly stronger momentum. The effect depends on article tone, reverses in the long run, is more pronounced for stocks with high uncertainty, and is stronger in states with high investor individualism. Our findings suggest that media coverage can exacerbate investor biases, leading return predictability to be strongest for firms in the spotlight of public attention. These results collectively lend credibility to an overreaction-based explanation for the momentum effect. (*JEL* G12, G14)

The momentum effect is among the strongest and most pervasive return anomalies. Although its existence has been convincingly documented in different time periods, countries, indexes, and asset classes,[1] a central issue is still far from being resolved: What are the underlying causes of momentum?

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[1] The momentum effect was first described by Jegadeesh and Titman (1993). For U.S. stock momentum in earlier and later time periods see, for example, Chabot, Ghysels, and Jagannathan (2009) and Fama and French (2008), respectively. For international evidence, see, for instance, Asness, Moskowitz, and Pedersen (2013), Chui, Titman, and Wei (2010), Fama and French (2012), and Rouwenhorst (1998). For momentum at the stock index level, see, for example, Chan, Hameed, and Tong (2000) or Moskowitz, Ooi, and Pedersen (2012). Momentum effects in other asset classes have been identified by, for example, Asness, Moskowitz, and Pedersen (2013) or Erb and Harvey (2006).

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Why exactly do winner stocks of the recent past tend to outperform loser stocks of the recent past? The goal of this paper is to employ an extensive media data set to shed new light on this long-standing debate.

Recent research demonstrates that media coverage directly affects the way in which investors collect, process, and interpret information (e.g., Engelberg and Parsons 2011; Tetlock 2007). Collectively, findings suggest "a potentially important role for the media in shaping the behavior of the stock market" (Hong and Stein 2007, 118). The interesting link to the momentum literature lies in the fact that investors' attention and information processing also play a crucial role in prominent behavioral theories of momentum as we outline below.[2]

For our study, we rely on a novel and carefully constructed data set of newspaper articles. It comprises approximately 2.2 million news stories in four leading national, as well as forty-one local U.S. newspapers from 1989 to 2010, which we match to 7,815 firms. The essence of our findings is captured in Figure 1.

The cumulative profits for a winner minus loser long-short portfolio are displayed separately for stocks in the highest media coverage quintile (bright solid line) and lowest quintile (dark solid line). Media coverage is based on the number of firm-specific articles in the *New York Times*, the *USA Today*, the *Wall Street Journal*, and the *Washington Post*, published in the six months of the return formation period. Importantly, we rely on a residual from cross-sectional regressions that control for firm variables known to drive the probability of being covered (most notably firm size). We describe the methodology in detail in the next section, and the basic advantage is that we can better isolate the true impact of (excess) appearance in the media.

The plot reveals a clear and economically strong positive relation between residual media coverage and the size of initial momentum profits and subsequent reversal. The winner minus loser spread amounts to approximately 6.3% after six months in the high media coverage portfolio, but only to 2.1% in the low media coverage portfolio. The resulting 4.2% return difference closely corresponds to a monthly return spread of 68 bp that we obtain using the Jegadeesh and Titman (1993) portfolio construction procedure. The difference in momentum profits mostly can be attributed to the winner side; it is economically large (the unconditional momentum return over the sample period is just 65 bp per month); and it is highly statistically significant with a t-statistic of 3.7.

However, this media-based momentum is not a permanent effect. Ten months after portfolio formation, the reversal begins for both momentum portfolios, but it is substantially stronger in the high media coverage portfolio. As a

[2] The profitability of momentum is difficult to reconcile with rational asset pricing models. For instance, Chui, Titman, and Wei (2010, 362) note: "Given the magnitude of momentum profits, about 12% per year in the United States and Europe, they are unlikely to be explained by risk-based theories." Consistent with this view, most theoretical and empirical work focuses on behavioral approaches to explain momentum.

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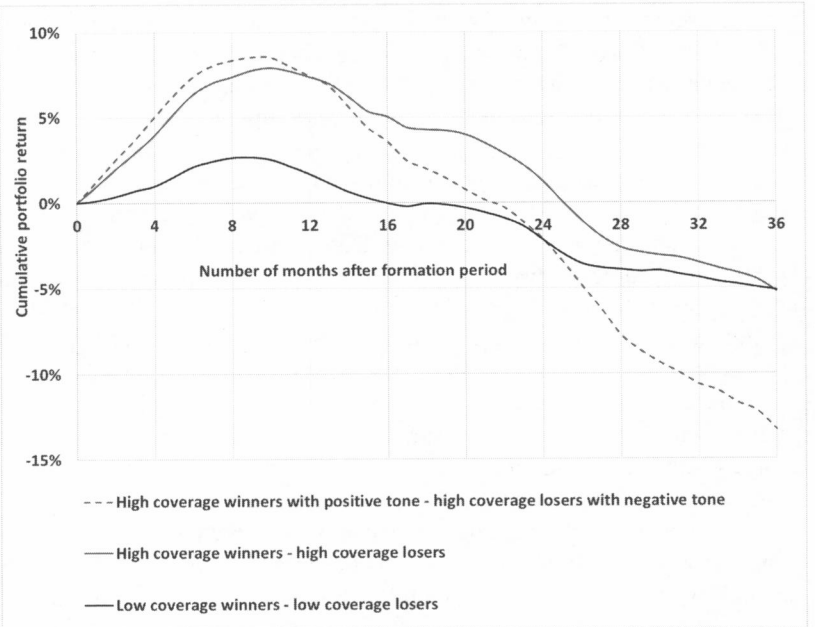

Figure 1
Buy-and-hold long-term momentum profits for firms with high and low residual media coverage
This figure shows the cumulative buy-and-hold returns for a winner minus loser long-short strategy separately for three different stock categories. In each month, residual stock-level media coverage is obtained from a month-to-month rolling OLS regression with ln (1+number of media articles) as dependent variable and size, analyst coverage, and dummies for S&P 500 and NASDAQ membership as explanatory variables. For the portfolio construction, stocks are first divided into residual media coverage quintiles (averaged over the formation period) and then further separated into a loser portfolio (30% of stocks with the lowest return over the previous six-month formation period), a neutral portfolio, and a winner portfolio (30% of stocks with the highest return over the formation period). The first category (dark solid line) compiles stocks for the long and short portfolio of stocks in the lowest residual media coverage quintile. The second category (bright solid line) consists of stocks falling in the highest residual media coverage quintile. The third category (bright dashed line) is a subset of the stocks in the second category and additionally conditions on the tone of the press articles based on a median split in each portfolio. Tone is measured as the fraction of negative words based on the word list proposed by Loughran and McDonald (2011). To be consistent with our approach of measuring media coverage, the tone of the articles is also orthogonalized with respect to size, analyst coverage, S&P 500 membership, and NASDAQ membership in each month.

consequence, the cumulative return difference between both portfolios shrinks toward zero after 36 months.

The overall pattern in Figure 1 is very robust. Given the endogeneity of press articles, we are careful in making sure that we do not simply pick up a spurious correlation between media coverage and momentum. For instance, newspapers may be more attentive towards firms that did particularly well or poorly in the recent past. If the extremity of past returns was also systematically related to momentum profitability, then not controlling for this effect would lead to the incorrect conclusion that media coverage per se, and not its interaction with formation period returns, causes stronger momentum. As recently uncovered by Bandarchuk and Hilscher (2013), this line of reasoning indeed affects the

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role of several stock-level variables, such as analyst coverage or turnover, which previously have been argued to enhance momentum profits. Applying the same methodology, we find that media coverage is among the very few surviving characteristics and thus convincingly supports behavioral momentum explanations.

Our baseline findings of momentum and reversal effects that are (all else equal) more pronounced for firms with high media coverage are suggestive of an overreaction explanation in the spirit of Daniel, Hirshleifer, and Subrahmanyam (1998). The authors advocate a model in which momentum arises as a result of two central investor biases: overconfidence and self-attribution. Mistaken beliefs lead investors to overweight (underweight) subsequent public signals that confirm (contradict) their initial private information. Confirming news will be considered as evidence of one's own skills, whereas disconfirming news will be largely neglected. As a consequence, overconfidence increases even further and prices temporarily overshoot, before the mispricing is gradually corrected.

Coverage in popular newspapers catches investors attention (e.g., Engelberg and Parsons 2011; Solomon, Soltes, and Sosyura 2014) and also may be seen as an indicator that a firm is in the spotlight of public discussion. At the same time press articles in popular newspapers often contain rather vague, ambiguous, or simply stale information that is less value relevant (see, e.g., Gurun and Butler 2011; Huberman and Regev 2001; Solomon 2012; Tetlock 2011), and investors are likely to "rely on them to form their opinions" (Zingales 2000, 1,628). This suggests that media coverage (in contrast to, e.g., newswires) might be considered to be a good proxy for attention to stale news about a firm.

Moreover, individuals tend to be particularly overconfident and overreactive in settings in which more judgment is required to evaluate ambiguous information (see, e.g., Daniel and Titman 2006 and the references therein). Although the first signal in Daniel, Hirshleifer, and Subrahmanyam (1998) is referred to as "private," it really reflects intangible or vague information that could appear in public sources. Hence, in the context of the model, articles in newspapers measured during the momentum formation period might be best thought of as the first signal that acts as a confirmation device when later public, tangible informative signals arrive.

Byun, Lim, and Yun (2013), Cooper, Gutierrez, and Hameed (2004), Da, Engelberg, and Gao (2011), and Hou, Peng, and Xiong (2009) provide empirical support for the line of reasoning behind Daniel, Hirshleifer, and Subrahmanyam (1998). Our study differs from this work in two respects: First, we propose residual media coverage as a proxy related to the first signal in the model setting. Second, in addition to our baseline analysis, we design several conceptually diverse tests aimed at exploring how investor overconfidence, a central ingredient in the model, influences the magnitude of media-based momentum. In this in-depth analysis, we exploit unique features of our media data as an attention proxy.

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As such, we first evaluate the qualitative content of the articles using the dictionary method developed by Loughran and McDonald (2011). Our results, which are depicted in Figure 1 (bright dashed line), indicate that both media-based momentum and subsequent reversal are stronger among the subset of stocks for which the article tone matches the formation period return. This finding is consistent with the idea that an initial media-based signal for a winner stock might be seen as more (less) favorable if the tone is also particularly positive (negative). Second, we find that our results are roughly twice as large for stocks with high uncertainty as for stock with low uncertainty. Third, and motivated by the cross-country analysis of Chui, Titman, and Wei (2010), we use local newspaper coverage to document that media-based momentum tends to be stronger among U.S. states with higher individualistic tendencies.

Overall, substantial evidence indicates that the media's role on momentum profits can be explained by investor overreaction and that overconfidence appears to be a major driver behind this effect. In later parts of the paper, we discuss the extent to which alternative momentum theories, such as those of Lee and Swaminathan (2000) or Tetlock (2011), may be able to explain our results.

In addition to our contribution to the momentum literature, our paper is also linked to the emerging literature on the role of media in financial markets. Indeed, there is a controversial debate on whether media coverage makes capital allocations more efficient.

On the one hand, coverage has been argued to disseminate information to a broad audience (Fang and Peress 2009), to reduce information asymmetries (Tetlock 2010), and to potentially lead to faster diffusion of new information as shown for specific events (e.g., Huberman and Regev 2001; Peress 2008). This connects it to the gradual information diffusion model of Hong and Stein (1999), which argues that momentum is primarily caused by investor underreaction to fundamentally relevant information. The intuition behind the model has received empirical support by, e.g., Ang, Shtauber, and Tetlock (2013), Chan (2003), Da, Gurun, and Warachka (2014), Hong, Lim, and Stein (2000). Nevertheless, if one is willing to assume that excess media coverage, ceteris paribus, increases the speed of information dissemination, then we might expect momentum to be less (and not more) pronounced in high media coverage stocks.

In this respect, Chan (2003) is probably most close to our study. He analyzes differences in stock price reactions subsequent to large absolute returns in the previous month depending on whether the underlying company was mentioned in a headline or lead paragraph. He finds evidence of a strong drift without a reversal after bad news, which he interprets as evidence of slow information diffusion.

The Appendix provides a detailed discussion, but the major factors behind the partly different results in our study are as follows: First, his findings appear to be considerably driven by articles from newswires, which are likely to proxy

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for the arrival of valuable news. Second, the reported return drift is largely restricted to smaller and more illiquid stocks whereas we concentrate on larger, easily investable firms following Jegadeesh and Titman (2001). Third, there are important methodological differences. For instance, Chan (2003) relies on a binary analysis by quantifying whether or not a stock had coverage. In contrast, we compute an excess coverage measure that quantifies the unexpectedly high or low weight the media attaches to a given firm over a six-month period, holding important characteristics, such as firm size, fixed.

On the other hand, recent work has also identified a dark side of media coverage by arguing that it can induce temporary stock price distortions (e.g., Dougal et al. 2012; Engelberg, Sasseville, and Williams 2012; García 2013; Gurun and Butler 2011; Tetlock 2007). This stream of the literature suggests that popular press coverage may contribute to investor biases, such as overreaction to stale information.

Indeed, our results suggest that it is not always the "neglected corners" where the violation of the weak form of market efficiency is particularly pronounced. Instead, excess media coverage seems to exacerbate investor biases so that momentum and reversal effects can be strongest for firms in the spotlight of public attention.

1. Data and Empirical Setup

Our initial sample consists of all common stocks traded on NYSE, AMEX, or NASDAQ that appear on both CRSP and Compustat at any time during 1989 and 2010. To ensure that our findings are not driven by small illiquid stocks or bid-ask bounces, we follow Jegadeesh and Titman (2001) and exclude stocks with a market capitalization below the first NYSE size decile and stocks with prices below $5 at the end of the formation period. For all companies, we then collect newspaper articles from LexisNexis. Depending on the specific focus of the different tests in this study, we rely on major national newspapers, local newspapers, or both.

For the baseline examination, we follow Fang and Peress (2009) and focus on the four major U.S. newspapers with weekday circulation: *New York Times* (NYT), *USA Today* (USAT), *Wall Street Journal* (WSJ), and *Washington Post* (WP). To restrict ourselves to articles that truly address a specific company, we use the LexisNexis "relevance score" measure. In our baseline tests, we retain all articles with a relevance score from 80% to 99%. To gather all company-related articles, we search the LexisNexis database using company names from Compustat as key words in the company search function.[3]

[3] These are current names, which are appropriate in most cases because LexisNexis takes into account name changes. However, in few instances, articles before a name change were obviously missing. To obtain these articles, we also use historical names from CRSP (manually corrected for abbreviations) in the search engine and afterwards delete any duplicates. For all firms with more than 200 articles over the entire period, we conduct

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We additionally collect articles from the maximum set of local newspapers available via LexisNexis. A complete list of the resulting forty-one local and four national newspapers, as well as summary statistics, are given in Table 1.

In total, we are able to assign approximately 2.2 million newspaper articles to more than 7,800 different firms in our data set. About a quarter of these articles are released in one of the four major national newspapers, thereby covering more than 6,500 different firms. The Appendix shows summary statistics of national, local, and total media coverage. In general, media coverage is highly skewed. For instance, among firms that are covered by the national media in a given year, the mean number of articles is around 16, but the median is only three and the 25th percentile is one. We therefore define media coverage as ln(1+number of articles) in a given month.

Table 1 also displays the average tone in press articles by newspaper as quantified by textual analysis. We thereby follow the dictionary approach developed by Loughran and McDonald (2011), who evaluate the qualitative content of an article based on the fraction of negative words. The approach accounts for the finance jargon and is therefore particularly useful for our purposes (see also, e.g., García 2013). The *Wall Street Journal* contains the largest fraction of negative words (2.19%), appearing in line with conventional wisdom, as well as the findings of the few studies that have textually analyzed firm-specific newspaper articles so far (e.g., Gurun and Butler 2011; Tetlock, Saar-Tsechansky, and Macskassy 2008).

In Figure 2, we plot the time-series evolution of the percentage of firms with press coverage in consecutive six-month intervals, which match the length of the momentum formation period. As can be seen, coverage by all newspapers is relatively stable and fluctuates between about 45% and 65%. The sharp increase during the early 90s for local newspapers is largely a consequence of the limited availability in LexisNexis at the beginning of the sample period.

Next, we examine determinants of media coverage on the basis of cross-sectional Fama/MacBeth regressions in Table 2. The construction of the explanatory firm variables follows the standard in the literature and is explained in the Appendix.

A natural candidate is a firm's (log) market capitalization. Specification 1 shows that size is indeed a very important predictor, generating alone an adjusted R^2 of 24%. However, from specifications 2–4, which are also univariate, it turns out that S&P 500 membership, NASDAQ membership, and (log) analyst coverage have substantial explanatory power. Although the negative coefficient for the NASDAQ dummy is in line with the results of Fang and Peress (2009) and hence expected, the strong influence of the S&P 500 dummy with an R^2 of 18% is noteworthy. Specification 5 shows that, when additionally controlling for market capitalization, the influence of the S&P 500

additional plausibility checks to ensure that the articles really address the specific company and make corrections where necessary.

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Table 1
Summary statistics for the newspaper data set

Newspaper	Availability	Total no. articles	Mean no. articles per year	% of firms covered in given year	Average weekday circulation	Mean fraction of negative words
New York Times	**1989–2010**	**197,812**	**8,991**	**26.00%**	**1,100,174**	**1.82%**
Washington Post	**1989–2010**	**93,499**	**4,250**	**12.45%**	**716,553**	**1.95%**
Wall Street Journal	**1989–2010**	**217,374**	**9,881**	**37.51%**	**2,077,664**	**2.19%**
USA Today	**1989–2010**	**53,115**	**2,414**	**7.16%**	**2,192,879**	**1.86%**
Arkansas Democrat Gazette	1989–2010	33,595	1,679	6.49%	182,859	1.62%
Atlanta Journal and Constitution	1991–2010	74,196	3,710	10.99%	345,571	1.74%
Augusta Chronicle	1993–2010	46,641	2,587	8.08%	59,243	1.61%
Austin American Statesman	1994–2010	46,547	2,738	9.46%	173,850	1.64%
Birmingham News	1993–2010	19,076	1,060	6.18%	146,605	1.57%
Boston Herald	1992–2010	35,933	1,866	8.12%	205,057	1.91%
Buffalo News	1993–2010	43,915	2,405	9.12%	188,113	1.65%
Chicago Sun Times	1992–2010	89,351	4,703	13.13%	377,987	1.93%
Daily News New York	1995–2010	20,356	1,272	4.54%	674,668	1.92%
Dallas Morning News	1993–2010	90,053	4,932	12.87%	446,923	1.57%
Dayton Daily News	1994–2010	25,580	1,505	6.55%	125,376	1.59%
Denver Post	1994–2010	30,788	1,810	7.79%	322,449	1.73%
Fresno Bee	1994–2010	8,325	490	3.39%	157,422	1.44%
Houston Chronicle	1992–2010	75,188	3,945	11.71%	495,942	2.02%
Las Vegas Review Journal	1997–2010	9,464	667	3.93%	178,722	1.64%
Minneapolis Star Tribune	1992–2010	27,393	1,422	6.67%	353,366	1.67%
New Orleans Times Picayune	1993–2010	33,363	1,853	7.31%	214,053	1.72%
New York Post	1998–2010	28,374	2,164	7.28%	624,788	2.16%
Palm Beach Post	1989–2010	26,660	1,212	5.84%	168,693	1.77%
Pittsburgh Post Gazette	1993–2010	45,913	3,054	11.59%	224,973	1.98%
Richmond Times-Dispatch	1996–2010	26,908	1,779	8.41%	178,920	1.71%
Sacramento Bee	2002–2010	10,466	1,163	6.88%	281,962	1.76%
Salt Lake Tribune	1995–2010	21,496	1,264	6.94%	128,631	1.86%
San Antonio Express News	1996–2010	31,096	2,073	8.52%	225,712	1.68%
San Francisco Chronicle	1990–2010	48,618	2,282	8.45%	403,124	1.87%
San Jose Mercury News	1994–2010	75,183	4,423	10.81%	249,841	1.97%
Santa Fe New Mexican	1995–2010	3,731	232	2.30%	21,245	1.69%
Seattle Post Intelligencer	1990–2008	53,189	2,779	8.81%	139,818	1.92%
St. Louis Post Dispatch	1989–2010	96,368	4,380	13.30%	265,813	1.97%
St. Petersburg Times	1989–2010	56,279	2,558	9.58%	318,209	1.98%
Star Ledger (Newark, NJ)	1996–2010	35,888	2,760	10.48%	360,137	1.92%
Tampa Tribune	1998–2010	16,530	1,270	7.34%	223,510	1.62%
The Oklahoman	1992–2010	29,097	1,531	7.00%	189,827	1.35%
The Oregonian	1989–2010	49,670	2,258	7.33%	310,233	1.75%
The Philadelphia Inquirer	1994–2010	52,213	3,071	11.68%	338,394	1.95%
The Plain Dealer	1993–2010	61,085	3,358	10.17%	334,107	1.65%
The Providence Journal	1994–2010	31,923	1,878	8.78%	146,798	1.79%
The Record (Bergen)	1996–2009	50,682	3,568	12.97%	176,701	2.01%
The Virginian Pilot	1994–2010	17,674	1,040	6.37%	186,931	1.74%
Tulsa World	1996–2010	55,577	3,700	13.04%	134,668	1.79%
Wisconsin State Journal	1992–2010	19,649	1,034	6.15%	96,927	1.72%
Total number of articles		2,215,833				
Total number of articles in WSJ, WP, NYT, and USAT		561,800				

This table reports data on availability, total number of articles, average number of articles per year, the average percentage of firms covered in a given year, average weekday circulation, and the average fraction of negative words per article for each of the local newspapers and national newspapers (in bold). Articles are obtained from LexisNexis using the company search function and a "relevance score" of at least 80%. *Availability* is the maximum of the start of our sample period (1989) and the first year of complete journal coverage. To calculate the *average percentage of covered firms in a given year*, we first calculate the percentage of coverage separately for each year in which the newspaper is available. This number is based on all firms that appear in our sample at any point in time during that specific year. We then take the yearly average. *Average weekday circulation* is based on circulation data for the years 2004, 2006, and 2009, which were compiled by BurrellesLuce using the Audit Bureau of Circulations as a data source (see www.burrellesluce.com). For Augusta Chronicle and Santa Fe New Mexican, we obtain circulation data directly from the Audit Bureau of Circulations. *The mean fraction of negative words* in the newspaper articles is based on the word list proposed by Loughran and McDonald (2011).

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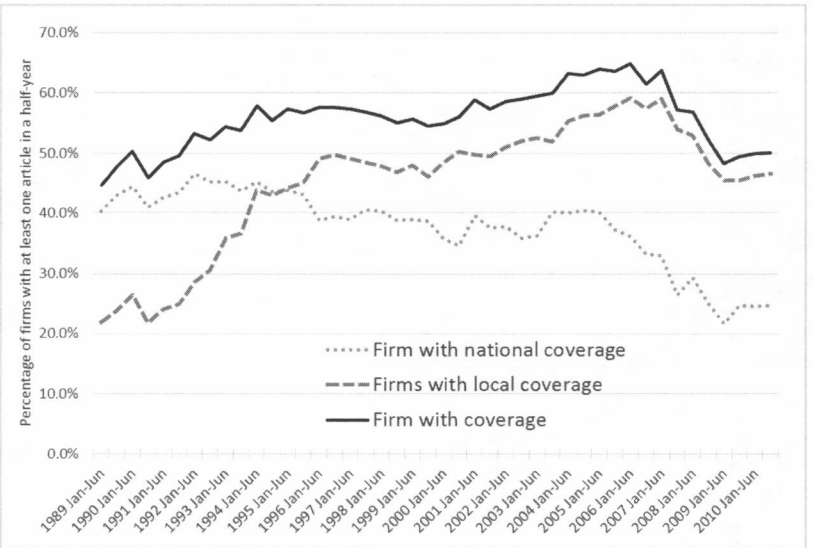

Figure 2
Percentage of firms with media coverage in six-month intervals
This figure shows the time-series evolution of the percentage of firms with press coverage separated by national, local, and all newspapers in a given half-year (January-June or July-December). The total media data set consists of forty-five different newspapers with weekday circulation, of which the four major U.S. newspapers *New York Times*, *USA Today*, *Wall Street Journal*, and *Washington Post* make up the national newspaper data set. For details about local newspapers, see Table 1.

dummy remains highly statistically significant. This suggests that S&P 500 membership might capture important nonlinearities between the amount of coverage and firm size, or alternatively that being a member of this popular index on its own increases the probability of receiving media attention.

In specifications 6 and 7, the NASDAQ dummy and analyst coverage are sequentially added to the model; that is, specification 7 looks as follows:

$$ln(1+no.\,articles)=\alpha+\beta_1\cdot ln(size)+\beta_2\cdot S\&P\ 500+\beta_3\cdot NASDAQ$$
$$+\beta_4\cdot ln(1+analyst)+\varepsilon_{media}. \quad (1)$$

Again in line with Fang and Peress (2009), analyst coverage has a negative impact on media coverage in this specification. This suggests that after controlling for firm size and index effects, analyst coverage and media coverage might be substitutes rather than complements. However, the model's adjusted R^2 of 26% implies a rather modest improvement in the overall explanatory power in comparison with the other specifications. Apparently, we approach the limits to explain media coverage. Indeed, the two final specifications 8 and 9 in Table 2 reveal that, when collectively adding further firm characteristics and also industry dummies, the additional gains in terms of explanatory power appear small with an overall adjusted R^2 of 30% in the rightmost column.

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Table 2
Multivariate regressions to explain media coverage

Model specification	Size 1	S&P 500 2	NASDAQ 3	Analyst 4	1 + S&P 500 5	5 + NASDAQ 6	Baseline model 7	7 + further controls 8	8 + industry dummies 9
Firm size	0.1907***				0.1467***	0.1490***	0.1603***	0.2073***	0.2084***
	(57.12)				(38.60)	(35.02)	(36.13)	(42.11)	(40.29)
S&P 500 dummy		0.6445***			0.2562***	0.2555***	0.2599***	0.2263***	0.2152***
		(62.37)			(22.44)	(23.05)	(23.13)	(20.93)	(21.61)
NASDAQ dummy			−0.2271***			0.0155***	0.0181***	−0.0149**	−0.0217***
			(−31.56)			(2.90)	(3.14)	(−2.14)	(−3.52)
Analyst coverage				0.2125***			−0.0275***	−0.0477***	−0.0477***
				(59.75)			(−19.19)	(−20.75)	(−25.03)
Book-market								0.0419***	0.0570***
								(10.21)	(10.67)
Turnover								−0.0074*	−0.0098***
								(−1.76)	(−2.65)
IVOL								0.1754***	0.1662***
								(21.65)	(17.94)
Mom strength								0.0370***	0.0385***
								(9.20)	(9.69)
No. of observations	264	264	264	264	264	264	264	264	264
Fama-French 48 dummies	no	no	no	no	no	no	no	no	yes
Adjusted R^2	0.2395	0.1821	0.0405	0.1092	0.2557	0.2566	0.2575	0.2787	0.2978

This table presents the results of various month-to-month OLS regressions to explain raw media coverage defined as *ln(1+no. articles)*. Specification 7 is our baseline model to obtain residual media coverage. *Size* is the natural log of market capitalization. *S&P 500* and *NASDAQ* are dummy variables that take the value of one when the firm is a member of the S&P 500 index or is listed on the NASDAQ. *Analyst coverage* is the natural log of (1+no. earnings estimates). *Book-market* is the firm's equity book-to-market ratio and *turnover* is the natural log of average share volume divided by shares outstanding using daily data over the last six months. *IVOL* (idiosyncratic volatility based on the Fama/French 3-factor model) and *Mom strength* are measured following Bandarchuk and Hilscher (2013). Further construction details are in the Appendix. The sample period covers M1:1989–M12:2010. Following Fama and MacBeth (1973), coefficients are calculated as time-series averages of monthly estimates, and t-statistics (in parentheses) are based on the time-series average and standard deviation. We adjust t-statistics for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

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The regression results suggest that we need to control for firm characteristics, most notably size, to isolate the true impact of media coverage on momentum profits. To do so, we closely follow the approach of Hong, Lim, and Stein (2000), who focus on residual analyst coverage, and who use firm size and NASDAQ membership as explanatory variables. We add an S&P 500 dummy due to its strong explanatory power for unconditional media coverage. In other words, we rely on the residuals of Equation (1) as our baseline measure of excess media coverage. In our view, this specification has the advantage of being still simple and easy to follow, while controlling for the key result of Hong, Lim, and Stein (2000) and removing the most serious dependencies between media coverage and further variables. In later checks, we verify the robustness of our findings using either simpler or more comprehensive models.

For now, we first take a closer look at the resulting distribution of various firm characteristics in Table 3 when sorting stocks into quintiles on the basis of raw media coverage (panel A) and residual media coverage (panel B).

As can be seen, using raw media coverage leads to large differences in firm characteristics across the five media portfolios. For instance, firms in the high coverage portfolio are more than ten times larger than firms in the low coverage portfolio and they are about six times more likely to be a member of the S&P 500 index.

Panel B shows that differences in firm characteristics between the extreme portfolios 1 and 5 substantially decrease with our residual media variables. This is not only the case for the variables included in the regression, but also for the book-to-market ratio, stock price, the Amihud (2002) illiquidity ratio, and firm age.

Nevertheless, panel B also shows some evidence of nonlinearities between residual media coverage and firm characteristics. For instance, firms in the intermediate quintiles tend to be smaller on average than firms belonging to portfolio 1 or 5. However, our empirical tests focus on the extreme media quintiles for which differences in firm-level variables are relatively small. Moreover, we later run additional tests, such as computing benchmark-adjusted returns as in Daniel et al. (1997), to further control for any remaining disparities.

2. Empirical Results

2.1 Baseline analysis: The effect of media on momentum returns

Following the standard in the literature, we construct momentum portfolios using a formation and a holding period of six months each, and skipping one month in between. To interact media exposure with momentum profits, we first create five equal-sized stock portfolios based on residual national media coverage within the momentum formation period. Within each of these quintiles, the winner (loser) portfolio then consists of all stocks with a return above the 70th percentile (below the 30th percentile) during their formation period. These return cutoffs are used throughout all tests in the paper.

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Table 3
Firm characteristics for portfolios sorted by raw and residual media coverage

Quintile	NYSE decile	Size (ln)	S&P 500	NASDAQ	Analyst (ln)	Book-market	Turnover (ln)	IVOL	Mom strength	Price	Amihud (ln)	Age (ln)
Panel A: Quintile sorts based on raw media coverage												
1	4.67	13.34	0.13	0.47	1.81	0.52	−0.34	13.35	0.33	26.19	−4.72	4.84
2	5.28	13.66	0.20	0.40	1.97	0.51	−0.20	12.97	0.30	28.16	−5.24	4.92
3	5.98	14.02	0.30	0.33	2.13	0.49	−0.10	12.25	0.29	31.20	−5.72	5.04
4	6.97	14.59	0.46	0.23	2.37	0.47	−0.05	11.01	0.27	67.22	−6.41	5.24
5	8.60	15.87	0.74	0.16	2.79	0.43	−0.03	9.18	0.24	109.50	−7.81	5.61
5-1	3.93***	2.53***	0.61***	−0.31***	0.98***	−0.08***	0.31*	−4.17***	−0.09**	83.30***	−3.09***	0.77***
t-stat 5-1	31.99	10.88	30.55	−20.99	20.05	−2.84	1.75	−20.36	−2.28	7.10	−6.97	14.58
Panel B: Quintile sorts based on residual media coverage												
1	8.20	15.24	0.71	0.19	2.62	0.44	−0.20	7.93	0.19	64.22	−7.04	5.54
2	6.08	14.10	0.21	0.34	2.11	0.47	−0.22	11.37	0.25	39.70	−5.71	5.03
3	4.52	13.34	0.12	0.46	1.82	0.52	−0.25	13.86	0.28	32.40	−4.80	4.82
4	4.74	13.45	0.18	0.42	1.90	0.54	−0.17	14.16	0.31	49.61	−5.01	4.85
5	7.26	15.06	0.51	0.26	2.48	0.46	−0.01	11.49	0.29	70.25	−6.93	5.31
5-1	−0.94***	−0.18	−0.20***	0.08***	−0.14***	0.02	0.20	3.56***	0.09***	6.03	0.11	−0.22***
t-stat 5-1	−6.39	−0.74	−5.59	6.19	−4.31	0.50	1.13	19.16	3.31	0.38	0.21	−4.68

This table presents time-series averages of the yearly mean of firm variables for quintile portfolios based on *raw media coverage* (panel A) and *residual media coverage* (panel B). *Raw media coverage* is defined as *ln(1+no. articles)*, *residual media coverage* is obtained from rolling month-to-month OLS regressions with *firm size*, *S&P 500*, *NASDAQ*, and *analyst coverage* as explanatory variables. *Firm size* is the natural log of market capitalization. *S&P 500* and *NASDAQ* are dummy variables that take the value of one when the firm is a member of the S&P 500 index or is listed on the NASDAQ. *Analyst coverage* is the natural log of (1+no. earnings estimates). *NYSE decile* refers to the firm's market capitalization decile at which breakpoints are based on NYSE stocks. Ten denotes the portfolio with the largest stocks. *Book-market* is the firm's equity book-to-market ratio, and *Turnover* is the natural log of average share volume divided by shares outstanding using daily data over the last six months. *IVOL* (idiosyncratic volatility based on the Fama/French 3-factor model) and *Mom strength* are measured following Bandarchuk and Hilscher (2013). *Price* is the share price in CRSP; *Amihud* is the Amihud (2002) illiquidity ratio; and *Age* is based on the first appearance of the stock's permco in CRSP. Further construction details are in the Appendix. The sample period covers M1:1989–M12:2010. The final rows in each panel show *t*-statistics associated with the (5)-(1) portfolio differences. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

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To quantify residual media coverage in this setting, we run the regression specified in Equation (1) separately for each of the six months of the formation period and then use the average residual obtained in these six cross-sectional regressions. In this step, firm months without coverage are also considered. However, when creating the five equally sized residual media coverage portfolios, only firms with at least one article over the formation period are taken into account. This restriction is motivated by our additional analysis of the qualitative content of articles in later sections. Conditioning on covered firms allows us to rely on the same samples in both tests.

The baseline effect of media coverage on momentum profits is displayed in Table 4. Returns are given in % per month (as in the remainder of the paper) and are based on overlapping equally weighted portfolios as in Jegadeesh and Titman (1993). t-statistics are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months.

As shown in panel A, there is a clear relation between media portfolios and momentum profits. Although the winner minus loser spread amounts to 33 bp per month (t-stat 1.42) for the lowest coverage portfolio and is only slightly larger in portfolio 3 (43 bp, t-stat 1.78), it is more than three times as large for the highest coverage portfolio (102 bp, t-stat 3.61). The difference of 68 bp per month (or more than 8% per year) is not only economically meaningful, but also statistically highly significant with a t-value of 3.70. In the following, we refer to this finding as the media-based momentum effect. The Appendix displays its full return distribution, as well as its time series. It also shows that media-based momentum, which is the difference between two momentum strategies, does not suffer from the type of momentum crashes recently uncovered by Daniel and Moskowitz (2013). In fact, it is slightly positively skewed.

Differences in the level of press coverage are large. As Table 4 uncovers, the mean number of articles in residual media coverage portfolios 1 to 5 are 2.50, 2.60, 2.71, 4.98, and 44.55. A very similar pattern is found for the median. In fact, the distribution of articles across portfolios is very similar to the one obtained using raw media coverage even though firm characteristics across portfolios are now comparable.[4]

If residual media coverage mattered for momentum returns, one would consequently expect particularly strong differences between portfolio 3 and 4, as well as between portfolio 4 and 5. This is indeed what we find. Despite this disproportionate increase in momentum profits along the media quintiles, the recently proposed monotonicity test of Patton and Timmermann (2010) verifies that momentum profits are strictly increasing with excess media coverage.

Notably, and in contrast to Hong, Lim, and Stein (2000), rising momentum profits are not primarily driven by the short leg of the portfolio. In fact, and

[4] Moreover, as Table 3 indicates, firms in portfolios 1 and 5 in general tend to be relatively large, liquid, and covered by many analysts. Untabulated findings show that there are no marked differences in these firm characteristics between the winner and the loser portfolio.

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Table 4
Residual media coverage and momentum: Baseline results

Residual media coverage portfolio	Loser return	Mid return	Winner return	Mom return	Mom t-stat	Mean / median no. of articles
Panel A: Double sorts and raw returns						
1	0.73	1.02	1.07	0.33	(1.42)	2.50 / 1.92
2	0.66	1.01	1.10	0.44*	(1.72)	2.60 / 1.34
3	0.74	1.00	1.17	0.43*	(1.78)	2.71 / 1.60
4	0.65	1.08	1.35	0.70***	(2.72)	4.98 / 3.30
5	0.54	1.06	1.55	1.02***	(3.61)	44.55 / 20.96
Return 5-1	−0.20	0.03	0.48***	0.68***		
t-stat 5-1	(−1.30)	(0.29)	(2.86)	(3.70)		
Panel B: Risk-adjusted returns for media-based momentum						
Factor model		1F	3F	4F	6F	
Intercept		0.61***	0.69***	0.64***	0.66***	
Intercept t-stat		(3.20)	(3.87)	(3.65)	(3.68)	
Adjusted R^2		0.04	0.12	0.12	0.12	
Panel C: Benchmark-adjusted returns for media-based momentum						
Benchmark characteristics		size + book-market	size+ turnover	size+ industry	DGTW	
Return		0.49***	0.54***	0.64***	0.42***	
t-stat		(2.70)	(3.01)	(3.58)	(2.69)	

This table presents momentum returns for stock portfolios sorted first by *residual media coverage*. To construct the momentum portfolios, we use a formation and holding period of six months each and skip one month in between. Residual media coverage quintiles are based on the firms' average residual media coverage from our baseline regression model during the formation period and consider only firms with at least one article over this period. Within each quintile, the *winner* (*loser*) portfolio consists of all stocks with a formation period return above the 70th percentile (below the 30th percentile). Momentum returns (reported in % per month) are based on overlapping portfolios that are equally weighted as in Jegadeesh and Titman (1993). Panel A shows raw returns; panel B shows risk-adjusted returns; and panel C characteristic-based benchmark-adjusted returns. Results in panels B and C are only for the media-based momentum effect, that is, the difference between the momentum return for the high and low residual media coverage portfolios. *1F* refers to the CAPM; *3F* refers to the Fama and French (1993) model; *4F* refers to the Carhart (1997) model; and *6F* refers to a model augmented with factors for short-term and long-term reversal. For benchmark-adjustment we use dependent sorts and subtract value-weighted benchmark returns as in Daniel et al. (1997). Industry-adjustment for the model *size+industry* is based on the first digit SIC code and excludes stocks with SIC Codes 0 or 9. *DGTW* refers to the original adjustment in Daniel et al. (1997). The sample period covers M1:1989–M12:2010. t-statistics (in parentheses) are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

in line with the reasoning in, for example, D'Avolio (2002) on short-selling constraints and Barber and Odean (2008) on attention effects in investor behavior, the winner portfolio makes a stronger contribution than the loser portfolio (48 vs. −20 bp, t-stat 2.86 vs. −1.30). Moreover, media coverage has only predictive power for firms contained in the long or short leg of a momentum strategy. For firms between formation period return percentile 30 and 70, the impact of the media is negligible.

As a further note, a first look at panel A suggests that, in contrast to the evidence reported by Fang and Peress (2009), low media coverage stocks do not earn higher returns than do high media coverage stocks. Specifically, the average return (added across all momentum portfolios) is $(0.73+1.02+1.07)/3=0.94\%$ for the low-media portfolio and $(0.54+1.06+1.55)/3=1.05\%$ for the high-media portfolio. However, we have verified that we can closely

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replicate the findings of Fang and Peress (2009) using their sample and methodology.[5]

Table 4 also examines the size of the media-based momentum effect on the basis of factor regressions. More specifically, the alphas reported in panel B result from a 1-factor model, a Fama and French (1993) 3-factor model, a Carhart (1997) 4-factor model, and a 6-factor model augmented with the short-term and long-term reversal factor. The alphas range from 61 to 69 bp and are thus virtually unchanged from the baseline effect.

Finally, although the regressions are helpful in determining whether the strategy loads on certain factors, they might be less suited in controlling for potential nonlinear relations between firm variables and residual media coverage. From Table 3, we know that such nonlinearities exist. To address these concerns, we start by following Daniel et al.'s (1997) procedure and benchmark-adjust stock returns before (re)calculating the media-based momentum effect. Doing so, we use the traditional characteristic-based benchmarks of Daniel et al. (1997), who sort stocks according to size quintiles, book-to-market quintiles, and prior return quintiles (denoted "DGTW").

In addition, we employ alternative two-dimensional benchmarks that sort firms according to size deciles and book-to-market deciles, size deciles and turnover deciles, and size deciles and industry membership. The use of these alternative benchmarks is motivated by the fact that the role of the later variables is commonly examined in the momentum literature (e.g., Hong, Lim, and Stein 2000; Lee and Swaminathan 2000; Moskowitz and Grinblatt 1999).

Panel C shows that the baseline effect is somewhat lower and ranges between 42 (traditional DGTW benchmarks) and 64 bp (size/industry adjustment). For all benchmark models, however, the spread is still highly statistically significant at the 1% level. These findings give us further assurance that media-based momentum is not explained by other factors.

2.2 Baseline analysis: Robustness tests

We now turn to further and more extensive robustness tests that change our empirical design in several dimensions. The main results are presented in Table 5.

For the sake of completeness, specification (1) uses raw (instead of residual) media coverage. With a 6-factor alpha of 24 bp (t-stat 1.34), media-based momentum is not significant. This is not surprising for two reasons. First, and as Table 3 shows in detail, sorting stocks on raw coverage is essentially a hidden sort on firm size. The mean NYSE size decile in the low (high) coverage portfolio is 4.67 (8.60). The difference in median values is even larger. At the

[5] We find the difference in returns to be mainly driven by differences in the sample coverage (whole CRSP/Compustat common stock universe versus NYSE stocks and 500 randomly selected NASDAQ stocks), the price filter, and the size filter. Please note that our approach in Table 4 also differs in other respects from that of Fang and Peress (2009): our sample period is about twice as long; we measure coverage over the whole momentum formation period; we exclude stocks without media coverage; and we rely on a residual measure.

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Table 5
(Residual) media coverage and momentum: Robustness results

Robustness specification	Raw difference	1F alpha	3F alpha	4F alpha	6F alpha
1. Using raw coverage	0.15	0.20	0.23	0.22	0.24
	(0.91)	(1.23)	(1.42)	(1.29)	(1.34)
2. Residual coverage (firm size only)	0.40**	0.30	0.42**	0.47**	0.48**
	(1.98)	(1.47)	(2.25)	(2.47)	(2.51)
3. Residual coverage (S&P 500 dummy only)	0.72***	0.71***	0.75***	0.64***	0.66***
	(4.37)	(4.30)	(4.66)	(3.96)	(4.14)
4. Residual coverage	0.65***	0.57***	0.66***	0.63***	0.64***
(firm size and S&P 500 dummy only)	(3.46)	(2.89)	(3.68)	(3.47)	(3.49)
5. Baseline model + size decile dummies	0.53***	0.50***	0.60***	0.60***	0.63***
	(2.95)	(2.60)	(3.57)	(3.58)	(3.65)
6. Baseline model + industry dummies	0.63***	0.55***	0.62***	0.56***	0.57***
	(3.44)	(3.01)	(3.42)	(3.11)	(3.11)
7. Baseline model + turnover	0.66***	0.59***	0.66***	0.63***	0.65***
	(3.56)	(3.01)	(3.69)	(3.49)	(3.54)
8. National and local media	0.56***	0.45**	0.52***	0.57***	0.58***
with region dummies	(3.45)	(2.55)	(3.09)	(3.48)	(3.43)
9. Reverse-dependent sort	0.57***	0.51***	0.56***	0.59***	0.61***
	(3.37)	(2.87)	(3.46)	(3.60)	(3.69)
10. Total no. articles during formation period	0.67***	0.66***	0.71***	0.60***	0.63***
	(3.47)	(3.27)	(3.77)	(3.32)	(3.37)
11. Panel regression with firm fixed effects	0.32*	0.29	0.35**	0.39**	0.41**
	(1.95)	(1.63)	(2.20)	(2.34)	(2.43)
12. Subperiod M1:1989–M12:1999	0.95***	0.73***	0.78***	0.64**	0.65***
	(3.43)	(3.00)	(3.75)	(2.50)	(2.74)
13. Subperiod M1:2000–M12:2010	0.42*	0.41*	0.40**	0.40**	0.38*
	(1.77)	(1.66)	(1.98)	(1.97)	(1.82)

This table presents the results of various robustness tests. In (1), we form portfolios based on raw media coverage. In all following models, we rely on a residual media coverage measure from various regression specifications. In (2), the residual is based on ln(firm size) as the only explanatory variable. In (3), the residual is based on a S&P 500 dummy as the only explanatory variable. Model (4) contains both firm size and a S&P 500 dummy. In (5), we augment our baseline model of residual media coverage (see Equation (1)) with dummies for size deciles. In (6), we include dummies for the forty-eight Fama/French industries. In (7), we include turnover. In (8), we use both local and national newspaper reports and control for regional differences with state dummies. Model (9) reverses the sorting procedure by first conditioning on formation period returns. In (10), we compute residual media coverage using only one regression based on the total number of articles over the whole formation period from t-6 to t-1. In (11), the residual is based on a panel regression with firm fixed and month fixed effects. Finally, (12) and (13) report the results for two subperiods of the entire sample (M1:1989–M12:1999 and M1:2000–M12:2010). The variable of interest is the difference in momentum returns (top 30% winner portfolio minus bottom 30% loser portfolio) between the highest and the lowest (residual) media coverage quintile. Momentum returns reported (in % per month) are based on overlapping portfolios that are equally weighted as in Jegadeesh and Titman (1993). *1F* refers to the CAPM; *3F* refers to the Fama and French (1993) model; *4F* refers to the Carhart (1997) model; and *6F* refers to a model augmented with factors for short-term and long-term reversal. t-statistics (in parentheses) are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

same time, standard momentum is significantly stronger among smaller firms than among the largest firms (see the Appendix and also, e.g., Hong, Lim, and Stein 2000; Fama and French 2008). This is probably in part due to the fact that firm size is also highly correlated with frictions and impediments to arbitrage, such as liquidity, spreads, and short-selling constraints (e.g., Nagel 2005). Consequently, this pronounced size effect runs counter raw media-based momentum, which is the difference between momentum among highly covered (and thus typically very large) and lowly covered (and thus typically small) firms.

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Second, attention is not necessarily monotonically related to the raw number of media articles and therefore also not necessarily monotonic in firm size. It appears reasonable to assume that investors would, all else equal, pay less attention towards the marginal newspaper article about a large firm for which such stories are commonplace than towards the marginal article about a smaller firm that is typically rather neglected. Thus, a residual that captures the "excess" amount of media coverage relative to other firms with similar characteristics might be a better proxy for the amount of (intangible) information that is being consciously processed by investors.

The bottom line is that, although it is certainly interesting to explore how momentum varies with raw coverage, this probably does not constitute a clean test of the isolated impact of media coverage on momentum. To address these concerns in an intuitive way, we consider size-adjusted media coverage, which is simply the residual from regressing raw coverage either on firm size alone (2) or on a S&P 500 dummy alone (3). With 6-factor alphas of 48 (t-stat 2.51) and 66 bp (t-stat 4.14), media-based momentum is large and significant. Another parsimonious model (4) that includes both firm size and S&P 500 membership to account for nonlinearities in the typical amount of coverage leads to similar insights.

Next, we consider more comprehensive models of residual media coverage by including additional factors in the OLS baseline regression of Equation (1). The selection of these variables is guided by remaining differences in firm characteristics as mostly indicated by Table 3. More precisely, we include dummies for size deciles (5), dummies for the 48 Fama/French industries (6), or turnover (7). Findings are similar to the ones obtained for our baseline model.

In specification 8, we add local newspaper articles. As local outlets tend to cover local stocks, certain geographic regions might receive systematically high or low residual media coverage values. To mitigate this issue, we include state dummies in the regression. Findings show that the inclusion of local media articles has no substantial impact on the results.

Differences are also rather minor if we reverse the dependent sorting procedure for media coverage and formation returns (9). In specification 10, we compute residual media coverage using only one regression based on the total number of articles over the whole period from t-6 to t-1 as opposed to running six regressions and computing the average residual. Findings are nearly identical to the baseline analysis. As an alternative econometric way to control for time-invariant heterogeneity in coverage at the firm level, we also run a panel regression with firm fixed and month fixed effects in (11). With a 6-factor alpha of 41 bp, findings are weaker but remain economically and statistically significant (t-stat 2.43).

Finally, specifications 12 and 13 show results for the two subperiods from January 1989 to December 1999, as well as from January 2000 to January 2010. Results appear persistent, though weaker in the more recent period (see also Tetlock 2011). However, momentum itself was by far less successful in the

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2000s with an average monthly spread between the winner and loser portfolio of just 23 bp (t-stat 0.48). Hence, the difference in momentum returns of about 40 bp that we obtain by conditioning on media coverage is economically still quite substantial.

The Appendix shows further sensitivity checks that verify that our baseline results are remarkably robust with respect to changes in the estimation of residual media coverage, changes in the sorting procedure, and changes in the eligible newspaper or stock data set.

However, we have not yet addressed the recent critique brought forward by Bandarchuk and Hilscher (2013). The authors revisit the literature on "enhanced momentum," which typically sorts stocks first by stock-level characteristics deemed to be proxy for behavioral or rational momentum drivers and then by past returns to consequently document higher momentum profits.

Their analysis reveals that, in most cases, it is not the characteristic per se that matters, but its interaction pattern with extreme past returns. Adequately controlling for this relationship with a suitable regression approach causes the enhanced momentum profits seemingly stemming from, for example, turnover, analyst coverage, or the book-to-market ratio, to vanish almost entirely.

To address these findings, we imitate their Fama/MacBeth regression-based framework (see also Fama and French 2008). Momentum profits are defined as a stock's forward return above (below) the median, multiplied by a winner/loser dummy, which takes on a value of 1 (-1) if the stock is a winner (loser) in the formation period. The dependent variable is either the momentum profit in $t+1$, as in Bandarchuk and Hilscher (2013), or the average momentum return over a six-month period, which better corresponds to our baseline approach.

To infer to which extent extreme returns over the formation period cause high momentum profits, we calculate two formation period control variables, denoted as momentum strength and idiosyncratic volatility rank. Computational details are outlined in the description of Table 6. The independent variable of interest is residual media coverage, computed as in our baseline approach. In some specifications, we additionally include variables that previously have been linked to cross-sectional differences in momentum profits and that at the same time are not already taken into account by the baseline model on estimating residual coverage. More precisely, controls comprise the book-to-market ratio, turnover, firm age, the Amihud illiquidity ratio, nominal share price, and Fama/French 48 industry dummies.

We can univariately confirm the influence of residual media coverage on momentum returns (see specification 1). We also reconfirm the finding that momentum strength and idiosyncratic volatility rank have a positive impact on momentum (specification 2).

Importantly, when using all three variables as predictors in specification 3, we find that the coefficient of residual media coverage decreases only modestly and its level of statistical significance is not affected. Finally, in specifications 4 and 5, the impact of residual media coverage also survives the inclusion of additional

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Table 6
Momentum, residual coverage, and stock characteristics: Fama/MacBeth regressions

Panel A: Momentum return in t+1

Variable / model specification	1	2	3	4	5
Residual media coverage	0.1571***		0.1336***	0.1213***	0.0994**
	(3.74)		(3.47)	(2.96)	(2.40)
Momentum strength		0.3733*	0.3628*	0.3348	0.3634*
		(1.76)	(1.71)	(1.62)	(1.78)
Idiosyncratic volatility		0.0230***	0.0227***	0.0232***	0.0197***
		(4.63)	(4.60)	(2.98)	(2.92)
Firm characteristics	no	no	no	yes	yes
Industry controls	no	no	no	no	yes
Average adjusted R^2	0.00	0.01	0.01	0.02	0.03

Panel B: Average momentum return from t+1 to t+6

Residual media coverage	0.1228***		0.1136***	0.1031***	0.0785**
	(3.76)		(3.73)	(3.32)	(2.52)
Momentum strength		0.4338**	0.4248**	0.4062**	0.4160**
		(2.42)	(2.37)	(2.36)	(2.37)
Idiosyncratic volatility		0.0089***	0.0085**	0.0010	0.0049
		(2.67)	(2.55)	(0.21)	(0.93)
Firm characteristics	no	no	no	yes	yes
Industry controls	no	no	no	no	yes
Average adjusted R^2	0.00	0.01	0.01	0.02	0.04

Following Bandarchuk and Hilscher (2013), we run Fama/MacBeth regressions of momentum profits on stock characteristics, *idiosyncratic volatility*, and extreme past returns (denoted as *momentum strength*). Stock characteristics include *residual media coverage* (which is already orthogonalized with respect to firm size, S&P 500 membership, NASDAQ membership and analyst coverage) and a set of additional factors that have been shown to impact the magnitude of momentum returns in previous work (book-to-market, turnover, firm age, the Amihud (2002) illiquidity ratio, share price, 48 Fama/French industry dummies). *Residual media coverage* is computed as in Table 4. *Momentum strength* is calculated as exp(absolute difference between a stock's formation period log return and the median formation period log return of all stocks in the sample)-1. As in Bandarchuk and Hilscher (2013), we use log returns to achieve comparability in returns for extreme winners and extreme losers. *Idiosyncratic volatility* is the residual return from a month-to-month rolling Fama and French (1993) 3-factor regression using daily data. The residuals are then averaged over the six-month formation period. We sort stocks into twenty-five different volatility portfolios and use the resulting rank as independent variable (1-25). *Momentum return*, the dependent variable, is the stock's forward return minus the sample median return, multiplied with a dummy variable being 1 (-1) if the stock was a winner (loser). Regressions aim at predicting the momentum return in t+1 (panel A) and the average momentum return from t+1 to t+6 (panel B). The sample period covers M1:1989–M12:2010. t-statistics (in parentheses) are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

firm characteristics and industry controls. For the sake of brevity, coefficients for the firm characteristics are not tabulated. However, most coefficients are insignificant, which reconfirms the insights of Bandarchuk and Hilscher (2013).

3. Overconfidence-Driven Overreaction as Explanation for Media-Based Momentum?

In the following, we describe the main insights from additional tests designed to investigate possible drivers of media-based momentum in greater depth. One way to interpret the results is to argue that they fit neatly into the Daniel, Hirshleifer, and Subrahmanyam (1998) framework. Nevertheless, as we outline in the respective sections, at least some of the findings appear to be consistent

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with other models as well. In this sense, our evidence in support of a specific theory should be considered to be more circumstantial than definitive.

The analysis proceeds in four steps. First, we test whether medium-run media-based momentum is followed by a long-run reversal, which would be suggestive of investor overreaction. As the data lends credibility to this line of reasoning, we next explore whether we can enhance media-based momentum by conditioning on variables deemed to strengthen investor overconfidence in the spirit of Daniel, Hirshleifer, and Subrahmanyam (1998). More precisely, we consider article-level tone in a second step, firm-level uncertainty in a third step, and state-level individualism in a fourth step.

3.1 Long-run reversal

Tests based on long-run reversal help distinguish between overreaction-based versus underreaction-based explanations of momentum. We thus follow the previous literature in analyzing profits for up to 36 months after portfolio formation. Table 7 displays the main results. In this section, we focus only on the first four columns in this table, which depict the long-run performance of the baseline media-based momentum strategy. For the sake of brevity, Columns 5 to 8 already show the long-run results of media-based momentum strategies that additionally condition on article tone. These results will be discussed in the next section when we introduce details of the tone analysis.

Focusing on the baseline analysis, a clear pattern emerges, which is also evident from Figure 1, as previously discussed. Judging from a period of up to twelve months after formation as displayed in panel A, the high residual coverage momentum portfolio (still) outperforms the low coverage portfolio at the 5% significance level and by roughly 40 bp per month. After roughly ten months, however, the effect slowly starts to reverse. As panel B shows, in years 2 and 3 after portfolio formation, the high coverage momentum portfolio underperforms its low coverage counterpart by an average of 22 bp per month.

Panel C shows that the underperformance of the high residual coverage momentum portfolio in t+13 to t+36 completely offsets the higher initial momentum gains in the first twelve months. As can be seen, the difference between high and low coverage firms is virtually zero over the entire three-year period. The finding holds for both raw and risk-adjusted momentum returns.

The strong reversal indicates that investor overreaction appears to be the driving force behind media-based momentum, suggesting that our findings so far can best be reconciled with models such as those of DeLong et al. (1990) or Daniel, Hirshleifer, and Subrahmanyam (1998). Also Tetlock's (2011) stale-information theory may be particularly applicable for our results. To the extent that investors cause overreaction by trading on redundant information, high media coverage could proxy for the amount of stale news and overreaction "potential" in his model, even though the reversals do not occur within days, as occurs in his high-frequency analysis.

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Table 7
Medium-run momentum and long-term reversal

	Unconditional residual media coverage				Tone-enhanced residual media coverage			
	Loser return	Winner return	Momentum return	Momentum t-stat	Loser return	Winner return	Momentum return	Momentum t-stat
Panel A: Momentum effect in t+1 to t+12								
Res. coverage portfolio 1	0.85	0.97	0.12	(0.52)	0.96	0.94	−0.02	(−0.08)
Res. coverage portfolio 5	0.74	1.28	0.54**	(2.02)	0.83	1.29	0.46	(1.39)
Return 5-1	−0.11	0.31	0.42**		−0.12	0.35*	0.48**	
t-stat 5-1	(−0.39)	(1.58)	(2.42)		(−0.75)	(1.90)	(2.09)	
Factor model	1F	3F	4F	6F	1F	3F	4F	6F
Intercept	0.37**	0.46***	0.40**	0.40**	0.47*	0.56**	0.44*	0.42*
Intercept t-stat	(2.17)	(2.93)	(2.49)	(2.47)	(1.94)	(2.48)	(1.82)	(1.78)
Panel B: Momentum effect in t+13 to t+36								
Res. coverage portfolio 1	1.22	1.01	−0.21	(−1.58)	1.25	0.96	−0.29**	(−2.10)
Res. coverage portfolio 5	1.45	1.02	−0.43**	(−2.54)	1.62	0.97	−0.64***	(−3.44)
Return 5-1	0.23	0.01	−0.22		0.37**	0.01	−0.35**	
t-stat 5-1	(1.41)	(0.13)	(−1.63)		(2.12)	(0.08)	(−2.46)	
Factor model	1F	3F	4F	6F	1F	3F	4F	6F
Intercept	−0.23*	−0.13	−0.11	−0.13	−0.35**	−0.26**	−0.27**	−0.29**
Intercept t-stat	(−1.66)	(−1.38)	(−1.08)	(1.21)	(−2.36)	(−2.48)	(−2.46)	(−2.58)
Panel C: Momentum effect in t+1 to t+36								
Res. coverage portfolio 1	1.10	1.00	−0.11	(−0.85)	1.17	0.95	−0.21	(−0.85)
Res. coverage portfolio 5	1.23	1.13	−0.10	(−0.59)	1.37	1.09	−0.29	(−0.59)
Return 5-1	0.13	0.13	0.01		0.21	0.14	−0.08	
t-stat 5-1	(0.84)	(0.90)	(0.04)		(1.27)	(0.92)	(−0.50)	
Factor model	1F	3F	4F	6F	1F	3F	4F	6F
Intercept	−0.00	0.01	0.01	0.01	−0.01	0.00	−0.00	−0.01
Intercept t-stat	(−0.20)	(1.05)	(0.85)	(0.74)	(−0.45)	(0.19)	(−0.22)	(−0.39)

This table presents momentum returns over different holding periods for stock portfolios sorted on either *unconditional residual media coverage* (left-side columns) or *tone-enhanced residual media coverage* (right-side columns, past winners with positive tone and past losers with negative tone). See Tables 4 and 8 for a detailed description of how portfolios are constructed. Tested holding periods are from month t+1 to t+12 (panel A), from month t+13 to t+36 (panel B), and from month t+1 to t+36 (panel C) after portfolio formation. Momentum returns (reported in % per month) are based on overlapping portfolios that are equally weighted as in Jegadeesh and Titman (1993). The sample period covers M1:1989–M12:2010. In each panel, intercepts from the different factor models are reported only for the media-based momentum strategy, that is, the difference between the momentum return for the high and low residual media coverage portfolios. *1F* refers to the CAPM; *3F* refers to the Fama and French (1993) model; *4F* refers to the Carhart (1997) model; and *6F* refers to a model augmented with factors for short-term and long-term reversal. t-statistics (in parentheses) are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

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In contrast, given that return reversals completely offset media-based momentum in the long-run, our findings appear more difficult to reconcile with theories that attribute momentum at least partly to underreaction (e.g., Barberis, Shleifer, and Vishny 1998; Hong and Stein 1999).

Media coverage may also proxy for investor favoritism, which relates it to the momentum life-cycle hypothesis of Lee and Swaminathan (2000). For instance, reporters may be more attentive toward firms favored by the market because they wish to satisfy investors' demand for stories about these firms. This implies that high media coverage winners and losers are both in-favor stocks and conversely that low media coverage winners and losers are both out of favor. To the extent that this line of reasoning is true, the price continuations and reversals that we observe in relation to media coverage can be only partly explained by the model of Lee and Swaminathan (2000). For instance, according to the momentum life cycle, high media coverage winners, but not high coverage losers, ought to have particularly strong reversals. In contrast, we observe strong reversals particularly for high coverage losers.

3.2 Tone-enhanced media-based momentum

If overconfidence-driven overreaction as in Daniel, Hirshleifer, and Subrahmanyam (1998) was the driver behind media-based momentum, then we would expect that not only the intensity of media coverage mattered but also that its content mattered. Specifically, the effect should be particularly pronounced for those stocks for which the article tone matches the formation period return. For instance, an overconfident investor might consider its initial, media-based signal regarding a winner stock as being particularly (less) favorable if the tone of the article appears particularly positive (negative). As a consequence, conditioning on the content of press articles might enable us to predict future return patterns, both in the medium-run and in the long-run, even better.

Empirically, we start by adding a third sorting dimension to our baseline analysis. More precisely, after double-sorting stocks on residual media coverage and formation period return, we additionally perform a median split on article tone based on the approach developed by Loughran and McDonald (2011). To assure consistency with the way we measure media coverage, article tone is also orthogonalized with respect to the firm characteristics shown in Equation (1). Residual tone is estimated in a single regression based on the average tone of articles over the whole formation period. The Appendix confirms that we obtain similar results when relying on raw tone instead.

Panel A of Table 8 shows descriptive statistics for article tone. As one might expect, winner stocks in general have fewer negative words in their press articles than do loser stocks. However, the median split on tone also yields substantial variation within each (residual media coverage, momentum) portfolio.

We then replicate the baseline analysis of Section 2.1 separately for the subsample of stocks for which we expect high media-based momentum (positive tone winners, negative tone losers) and for the subsample for which

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Table 8
Tone-enhanced media-based momentum

Panel A: Percentage of negative words for different stock subgroups

Residual media coverage portfolio	Winner stocks		Loser stocks	
	negative tone	positive tone	negative tone	positive tone
1	2.73%	0.19%	3.33%	0.31%
5	2.60%	1.00%	3.04%	1.36%

Panel B: Media-based momentum with positive tone winners and negative tone losers

Residual media coverage portfolio	Loser return	Winner return	Momentum return	Momentum *t*-stat
1	0.81	1.13	0.31	(1.22)
2	0.76	1.07	0.32	(1.13)
3	0.84	1.22	0.38	(1.32)
4	0.70	1.29	0.59*	(1.95)
5	0.47	1.64	1.17***	(3.34)
Return 5-1	−0.34*	0.51**	0.86***	
t-stat 5-1	(−1.72)	(2.60)	(3.30)	
Factor model	1F	3F	4F	6F
Intercept	0.85***	0.94***	0.82***	0.82***
Intercept *t*-stat	(3.14)	(3.80)	(3.22)	(3.25)

Panel C: Media-based momentum with negative tone winners and positive tone losers

Residual media coverage portfolio	Loser return	Winner return	Momentum return	Momentum *t*-stat
1	0.66	1.01	0.35	(1.49)
2	0.57	1.13	0.56**	(2.04)
3	0.64	1.13	0.49**	(2.01)
4	0.59	1.41	0.81***	(3.26)
5	0.60	1.46	0.86***	(3.24)
Return 5-1	−0.06	0.45**	0.51***	
t-stat 5-1	(−0.40)	(2.45)	(2.64)	
Factor model	1F	3F	4F	6F
Intercept	0.37**	0.44**	0.48***	0.50***
Intercept *t*-stat	(2.05)	(2.45)	(2.65)	(2.75)

Panel D: Difference in media-based momentum: Panel B - panel C

Raw return	1F	3F	4F	6F
0.35	0.48*	0.50**	0.34	0.31
(1.31)	(1.88)	(2.07)	(1.31)	(1.23)

This table explores the role of article tone on media-based momentum. Tone is measured as the fraction of negative words based on the word list proposed by Loughran and McDonald (2011). Panel A shows the fraction of negative words in different momentum portfolios. Portfolios are constructed as in the baseline analysis (see Table 4), except that we add a third sorting dimension based on above and below median article tone within each portfolio. Panels B and C replicate the baseline analysis (see Table 4) conditioned on past winner and loser stocks with either negative or positive tone (see the panel description for details). To be consistent with our approach of measuring media coverage, the tone of the articles is first orthogonalized with respect to size, analyst coverage, S&P 500 membership, and NASDAQ membership. In each panel, intercepts from the different factor models are reported only for the tone-enhanced media-based momentum strategy, that is, the difference between the tone-enhanced momentum return for the high and low residual media coverage portfolios. *1F* refers to the CAPM; *3F* refers to the Fama and French (1993) model; *4F* refers to the Carhart (1997) model; and *6F* refers to a model augmented with factors for short-term and long-term reversal. *t*-statistics (in parentheses) are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

we expect low media-based momentum as the tone does not match (negative tone winners, positive tone losers). Table 8 depicts the main findings.

We indeed find that media-based momentum constructed from winners with positive tone and losers with negative tone is stronger (raw return: 86 bp,

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6-factor alpha: 82 bp) than in our baseline analysis in Section 2.1 (raw return: 68 bp, 6-factor alpha: 66 bp). As a consequence, the effect is considerably weaker (raw return: 51 bp, 6-factor alpha: 50 bp) in the subsample of stocks for which the tone does not match the formation period return. The Appendix confirms these insights using multivariate Fama/MacBeth regressions as in Table 6.

Depending on the specification, the annualized average return difference between tone-enhanced and tone-weakened media-based momentum is about 3.8% to 6% and thus is economically substantial. At the same time, there is a lot of variation in the time series, leading to statistical significance being achieved in two out of five models only. Arguably, while the Loughran and McDonald (2011) procedure to measure article tone can be considered as state of the art, any attempt to quantify article content is likely to be crude and sometimes incomplete, a point that is expressed by others as well (e.g., Dougal et al. 2012).

Nonetheless, consistent with a higher initial media-based momentum effect, the results reported in the rightmost columns of Table 7 also point towards a stronger reversal effect in years 2 and 3 after portfolio formation for the tone-enhanced media-based momentum strategy. For the period t+13 to t+36 (panel B), the reversal amounts to 35 bp per month (t-stat 2.46) as opposed to only 22 bp for the baseline strategy. In fact, the cumulative difference in returns over the entire three-year period after portfolio formation is -8 bp per month (panel C), which suggests that the initially higher momentum gains are more than offset by the subsequent reversal.

Overall, these return patterns appear closely in line with the implications of Daniel, Hirshleifer, and Subrahmanyam (1998). With regard to other momentum theories, the role of article tone is not unambiguously clear. For instance, it is challenging to assess whether positive (negative) articles about winning (losing) stocks are more or less likely to contain truly stale news, which would be particularly relevant for the setting in Tetlock (2011).[6]

Interpreting the idea of a momentum life cycle (as in Lee and Swaminathan 2000) very freely, one could also speculate that highly covered winner (loser) stocks with particularly positive (negative) tone might be hyped (excessively written off) by the media. In this way, they might classify as "late momentum stocks" for which the life-cycle theory predicts a high likelihood of long-term reversal. Although in line with our findings, this interpretation arguably differs from the way these "late momentum stocks" are empirically defined by Lee and Swaminathan (2000).[7]

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[6] If one is willing to assume that newspaper articles are generally stale, then one might expect a stronger overreaction and a subsequent reversal upon stale information that is presented as either highly positive or highly negative. Relative to a more neutral writing, these articles might potentially trigger more trading by irrational investors.

[7] More precisely, high trading volume winners and low trading volume losers are regarded as "late momentum stocks" by Lee and Swaminathan (2000). These stocks are considered to be particularly favored (in the case of

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Finally, the fact that the extent of the reversal is positively related to the extent to which the article tone matches the formation period return makes our findings seemingly less consistent with the idea of slow information diffusion (e.g., Hong and Stein 1999) as the primary driver of media-based momentum.

3.3 Uncertainty

Both theoretical and empirical work argues that investor overconfidence is likely to be positively related to the degree of uncertainty about a given stock (e.g., Daniel, Hirshleifer, and Subrahmanyam 1998; Daniel, Hirshleifer, and Subrahmanyam 2001; Zhang 2006). For instance, Jiang and Lee (2005) argue that solid feedback for young firms with high return volatility, high turnover, and high cash-flow duration is hard to obtain. Thus, learning is constrained and investors might regard their private signals as (all else equal) more plausible. As a consequence, overconfident investors are likely to trade particularly aggressively in firms whose businesses are, ceteris paribus, more difficult to value. This line of reasoning suggests that overreaction-driven momentum and reversal effects should be particularly pronounced in hard-to-value stocks which at the same time are often featured in the media.[8]

This hypothesis is not solely predicted by Daniel, Hirshleifer, and Subrahmanyam (1998). As uncertainty is likely to interact with a number of behavioral biases, stronger mispricings among hard-to-value stocks are in line with the implications of many mistaken-beliefs frameworks (see, e.g., Baker and Wurgler 2007 or Hirshleifer 2001 for discussions). Moreover, these stocks are often argued to be a natural habitat of less sophisticated investors (e.g., Barber and Odean 2008; Kumar 2009). As a consequence, also the model of Tetlock (2011), for example, could be consistent with stronger overreaction among stocks with high uncertainty.[9]

Empirically, we broadly follow Jiang and Lee (2005) in constructing an aggregate uncertainty proxy based on firm age, return volatility, turnover, and cash flow duration (see Table 9 for computational details).

We start by presenting portfolio-level findings from dependent triple sorts. As in our baseline approach (see Table 4), firms are sorted by residual media coverage and formation period returns. We then additionally distinguish between firms with above and below median uncertainty. Panel A of Table 9

winners) or neglected (in the case of losers) by investors. In contrast, stocks with high residual media coverage in general tend to have rather high turnover in our setting (see Table 3).

8 On the other hand, if the slow diffusion of valuable information was the primary mechanism behind our findings, then one might expect less momentum in hard-to-value stocks with high press coverage than in hard-to-value stocks with low press coverage.

9 A positive influence of idiosyncratic volatility (one of our uncertainty proxies) on momentum profits and subsequent reversals is also predicted by recent work of Vayanos and Woolley (2013), who propose that momentum is partly attributable to agency issues in institutional asset management. In this sense, stronger effects for stocks with high uncertainty might also be consistent with their model. However, the Appendix shows that media-based momentum does not seem to be strongly related to the fraction of institutional ownership, suggesting that such factors are not a major driver of our findings.

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Table 9
Uncertainty and media-based momentum

Panel A: Portfolio-level analysis

Stocks with low uncertainty	Loser return	Winner return	Momentum return	Momentum t-stat
Residual media coverage portfolio 1	0.94	1.03	0.09	(0.45)
Residual media coverage portfolio 5	0.83	1.36	0.53**	(2.11)
Return 5-1	−0.11	0.33***	0.44***	
t-stat 5-1	(−0.84)	(2.71)	(3.01)	
Factor model	1F	3F	4F	6F
Intercept	0.43***	0.46***	0.36**	0.35**
Intercept t-stat	(2.79)	(3.27)	(2.26)	(2.14)
Stocks with high uncertainty	**Loser return**	**Winner return**	**Momentum return**	**Momentum t-stat**
Residual media coverage portfolio 1	0.65	1.13	0.48	(1.50)
Residual media coverage portfolio 5	0.28	1.68	1.40***	(3.62)
Return 5-1	−0.37	0.55***	0.92***	
t-stat 5-1	(−1.57)	(2.71)	(2.96)	
Factor model	1F	3F	4F	6F
Intercept	0.82**	0.93***	0.93***	0.96***
Intercept t-stat	(2.52)	(3.01)	(3.02)	(3.05)
Difference in media-based momentum: high uncertainty - low uncertainty				
Raw return	1F	3F	4F	6F
0.48	0.38	0.48	0.57*	0.61*
(1.60)	(1.23)	(1.59)	(1.84)	(1.88)

Panel B: Fama/MacBeth regressions of average return from t+1 to t+6

Residual media coverage	0.1117***	0.0971***	0.0789***
	(3.72)	(3.45)	(2.72)
Low uncertainty	−0.1182**	−0.1195**	−0.1189**
	(−2.17)	(−2.24)	(−2.30)
High uncertainty	0.3149**	0.2341*	0.2450**
	(2.22)	(1.70)	(1.98)
Res. media x low uncertainty		0.0730	−0.0167
		(0.85)	(−0.16)
Res. media x high uncertainty		0.4955*	0.4849*
		(1.95)	(1.91)
Momentum strength	yes	yes	yes
Fama-French 48 industry dummies	no	no	yes

Panel C: Media-based momentum for different volatility periods

	High idio vola	Low idio vola	Difference
Raw return	0.99***	0.37**	0.62
Raw return t-stat	(2.93)	(2.14)	(1.57)
6F intercept	1.00***	0.31*	0.69*
6F intercept t-stat	(3.10)	(1.93)	(1.80)

This table explores the role of uncertainty for media-based momentum. In each month, we rank stocks independently with regard to idiosyncratic volatility over the formation period, turnover over the formation period, cash-flow duration computed as in Dechow, Sloan, and Soliman (2004), and age at the end of the formation period. In panel A, *uncertainty* is defined as percentile(idiosyncratic volatility)+percentile(turnover)+percentile(cash-flow duration)-percentile(age). The analysis is then performed analogously to our baseline analysis (see Table 4), except that we add a third dependent sorting dimension based on above or below median uncertainty in a given month. The intercepts for the different factor models are reported only for the "uncertainty"-enhanced media-based momentum strategies, that is, the difference between the momentum return for the high and low residual media coverage portfolios separately for the low and high uncertainty environment, respectively. *1F* refers to the CAPM; *3F* refers to the Fama and French (1993) model; *4F* refers to the Carhart (1997) model; and *6F* refers to a model augmented with factors for short-term and long-term reversal. In panel B, we create a *high uncertainty dummy* that is one if a firm is above percentile 66 with regard to idiosyncratic volatility, turnover, and cash flow duration, as well as in the bottom third with regard to firm age. A *dummy for low uncertainty* is defined analogously. The analysis in panel B is then performed as in Table 6. In panel C, we first rely on daily data and a Fama and French (1993) model to compute average stock-level idiosyncratic volatility based on all NYSE/AMEX/NASDAQ common stocks (share code 10 and 11) in a given month. We then use a median split to distinguish between periods of *high and low idiosyncratic volatility* and proceed to report raw returns, as well as 6-factor alphas associated with the baseline media-based momentum effect, separately for these two periods. * indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

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verifies that media-based momentum is indeed much higher in hard-to-value stocks (raw long-short return: 92 bp per month, 6-factor alpha: 96 bp) than in stocks that are easier to value (44 and 35 bp, respectively). The large economic difference in 6-factor alphas is also statistically significant.

As an alternative approach, we perform monthly firm-level Fama/MacBeth regressions similar to Bandarchuk and Hilscher (2013) and Table 6. We create separate dummies for high and low uncertainty firms similarly to Jiang and Lee (2005) and interact them with residual media coverage. Panel B shows that our findings confirm those of Jiang and Lee (2005) and Zhang (2006), in that the traditional momentum effect is stronger in hard-to-value firms.

However, and similar to those in panel A, our results also reveal an additional exacerbating role of media coverage: the more firms with high uncertainty are featured in the media, the stronger their momentum effect is. This finding is both statistically significant and economically meaningful. A one-standard-deviation change in residual media coverage among hard-to-value stocks is estimated to increase monthly momentum returns by 25 bp, even if we control for momentum strength and industry effects.

As the Appendix verifies, this effect is again only a temporary phenomenon. In fact, the long-term reversal in uncertainty enhanced media-based momentum more than offsets the initial gains, clearly pointing to investor overreaction.

In panel C, we explore time-series implications of our cross-sectional findings. In times during which the valuation of the typical firm becomes harder than usual, investors might on average exhibit more pronounced biases, so that media-based momentum might become stronger. Broadly following Kumar (2009), and for each month separately, we compute average stock-level idiosyncratic volatility based on daily data. We then use a median split to distinguish between times of high and low overall valuation uncertainty. Panel C shows that media-based momentum is indeed about 60 bp stronger in high uncertainty months, even though it also remains statistically significant and economically meaningful in low uncertainty environments.

3.4 Cross-sectional variation in individualism

Chui, Titman, and Wei (2010) propose an elegant way to distinguish between competing explanations for the momentum effect. The authors rely on the individualism index of fifty countries developed by Hofstede (2001) to explain differences in momentum profits around the world. Building on Daniel, Hirshleifer, and Subrahmanyam (1998), as well as on psychological research, Chui, Titman, and Wei (2010) first establish a link between the degree of individualism, on the one hand, and overconfidence and self-attribution bias, on the other hand. For instance, people in individualistic cultures (e.g., the United States or the United Kingdom) tend to believe much more that their abilities are above average than do people in collectivistic cultures (such as Japan). This behavior, however, is a central characteristic of overconfidence.

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Given these insights, Chui, Titman, and Wei (2010) hypothesize and verify that individualism is strongly positively related to the profitability of country-specific momentum strategies around the world.

The authors end by highlighting the need for future research on the link between cultural differences and patterns in stock returns. We aim to progress on this front by transferring the idea from the cross-country to the within-country perspective. In this context, Vandello and Cohen (1999) propose a state-level individualism/collectivism index based on eight indicators related to personal, social, economic, religious, and political views and practices.[10] It is generally understood that absolute differences in cultural norms only within the United States might be less pronounced than in the international context. On the other hand, as Vandello and Cohen (1999) stress, "many of the problems of extraneous variation that make isolating cross-cultural differences challenging" (p. 290) can be mitigated, so that the information contained in the index should provide valuable insights and allow for useful individualism rankings.

Indeed, the index, which contains collectivism scores for fifty U.S. states, is widely used in the psychological literature. However, to our knowledge, we are the first to exploit the index to explain (geographic) differences in return anomalies. The Appendix displays state rankings of the U.S. collectivism index. It shows the intuitively appealing result that Southern states (particularly the Deep South) have greater collectivistic tendencies, whereas the Mountain West and the Great Plains tend to be rather individualistic. We follow the consensus in the literature and employ a firm's headquarter as a proxy for its location.

In a different line of research, many recent finance papers have demonstrated that investors, both retail and professional, exhibit a strong preference for local stocks (e.g., Coval and Moskowitz 1999; Coval and Moskowitz 2001; Grinblatt and Keloharju 2001; Huberman 2001; Seasholes and Zhu 2010). For our analysis, two aspects of this so-called local bias are especially important. First, recent findings highlight the role of (local) media in this context (e.g., Gurun and Butler 2011; Engelberg and Parsons 2011). Second, trading action of local investors has been shown to affect economic aggregates, such as the turnover of local stocks (e.g., Jacobs and Weber 2012; Loughran and Schultz 2005), their valuations (e.g., Hong, Kubik, and Stein 2008), and their returns (e.g., Kumar, Page, and Spalt 2011; Shive 2012).

Combining insights from the literature on individualism and local bias suggests the following: provided that local cultural norms indeed affect investment decisions, their impact on return patterns should be most easily identified in the behavior of local firms' stocks. Consequently, a testable

[10] Specifically, the index comprises the following items (partly reversed): percentage of people living alone, percentage of elderly people living alone, percentage of households with grandchildren, divorce-to-marriage ratio, percentage of people without religious affiliation, ratio of people carpooling to work to driving alone, percentage of Libertarian votes over the last four presidential elections, and percentage of self-employed people.

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hypothesis for our setting is that we expect the media-based momentum effect to be particularly strong (weak) in more individualistic (collectivistic) areas.[11]

A distinctive feature of our study is that we can base our analysis on more than 1.65 million news stories from a geographically widely dispersed cross-section of forty-one local newspapers. It appears crucial to take the role of local newspapers into account in a test based on cross-sectional differences in investor behavior along a geographical line. To explore the hypothesis outlined above, we first conduct portfolio-level tests as in our baseline analysis. The main idea here is to compute the media-based momentum effect separately for portfolios of firms located in individualistic and collectivistic areas and to compare the difference.

We only consider local newspapers and augment the baseline model to estimate residual media coverage with state dummies. These dummies are deemed to control for potential cross-sectional differences in firm characteristics and industry composition, as well as the number and availability of local newspapers. We consider the whole sample stock universe to run the regression, but then form separate portfolios for collectivistic and individualistic areas. Both areas are defined in a way that they each comprise about 30% of sample firms. Panel A of Table 10 shows the main findings.

Although there is a considerable noise due to the rather low number of firms in the portfolios at certain times, the media-based momentum effect in collectivistic regions is with 45 bp comparatively low and only marginally significant at the 10% level. In contrast, in individualistic states, the effect is 72 bp and significant at the 1% level. Depending on the factor model, the difference in the media-based momentum effect between individualistic and collectivistic states consequently amounts to about 25 to 30 bp per month. However, despite this economically meaningful effect, the difference is not statistically significant.

In panel B, we thus run multivariate Fama/MacBeth regressions as in Table 6 to better isolate the impact of individualism. Again, only individualistic (30%) and collectivistic (30%) firms are included and residual media coverage is estimated in a regression with state dummies. Of primary interest is the interaction term of residual media coverage and an individualism dummy, which is one (zero) for individualistic (collectivistic) regions. Firm-level controls include industry classification, momentum strength, idiosyncratic volatility, turnover, book-to-market ratio, age, illiquidity, and share price. In some specifications, we also add state-level controls by including mean firm characteristics and GDP growth.

The setting also allows us to further sharpen the analysis with regard to cross-sectional variation in local bias (see, e.g., Hong, Kubik, and Stein 2008;

[11] If overconfidence-based overreaction explained momentum, we would (in analogy to Chui, Titman, and Wei 2010) also expect that the traditional momentum effect without conditioning on media coverage was stronger in individualistic states. The Appendix provides evidence supporting this prediction.

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Table 10
Individualism and media-based momentum

Panel A: Portfolio-level analysis with state dummies

	Collectivistic areas		Individualistic areas	
Residual media coverage portfolios	Momentum return	Momentum t-stat	Momentum return	Momentum t-stat
1	0.38	(1.18)	0.19	(0.76)
5	0.83***	(3.14)	0.92***	(3.33)
Return 5-1	0.45*		0.72***	
t-stat 5-1	(1.66)		(2.87)	
Difference in media-based momentum: individualistic - collectivistic				
Raw	1F	3F	4F	6F
0.27	0.25	0.32	0.30	0.31
(0.83)	(1.13)	(1.01)	(0.94)	(1.00)

Panel B: Fama/MacBeth regressions

	Mom return in t+1		Mom return from t+1 to t+6	
	1	2	1	2
Momentum strength and idio vola	yes	yes	yes	yes
Firm characteristics	yes	yes	yes	yes
Fama-French 48 industry dummies	yes	yes	yes	yes
State-level controls	no	yes	no	yes
Baseline analysis				
Res. media x individualism dummy	0.1255	0.1385*	0.0628	0.0685
	(1.58)	(1.76)	(1.29)	(1.43)
Only states with above median local bias (based on retail and institutional investors)				
Res. media x individualism dummy	0.1411*	0.1357*	0.0576	0.0585
	(1.85)	(1.80)	(1.28)	(1.29)
Only states with above median local bias (based on retail investors only)				
Res. media x individualism dummy	0.141*	0.1347*	0.0622	0.0651
	(1.88)	(1.80)	(1.39)	(1.43)
Tone-enhanced analysis				
Res. media x individualism dummy	0.2421**	0.2507**	0.1248*	0.1282*
	(2.15)	(2.31)	(1.91)	(1.94)

This table compares media-based momentum in collectivistic and individualistic areas. Both collectivistic and individualistic areas are constructed in a way so that they each contain about 30% of sample firms. Only local newspapers are considered. Panel A displays portfolio momentum returns stemming from a double sort of residual media coverage (five portfolios in ascending order) and formation period returns. The model of residual coverage is the baseline model (see Table 4) augmented with state dummies. Momentum returns (reported in % per month) are based on overlapping portfolios that are equally weighted as in Jegadeesh and Titman (1993). t-statistics (in parentheses) are adjusted for serial autocorrelation using Newey and West (1987) standard errors with a lag of five months. See Section 3.4 for a more detailed description of this analysis. Panel B shows firm-level Fama/MacBeth regressions as in Table 6. Of primary interest is the *interaction term of residual media coverage* and an *individualism dummy*, which is one (zero) for individualistic (collectivistic) regions. Firm-level controls include industry classification, momentum strength, idiosyncratic volatility, turnover, book-to-market ratio, age, Amihud illiquidity, and share price. State-level controls include mean firm characteristics and (current and lagged) gdp growth. State-level estimates of local bias are taken from Korniotis and Kumar (2013). The tone-enhanced analysis broadly follows Table 8, but we orthogonalize the article tone with respect to state dummies (and size, analyst coverage, S&P 500 membership, and NASDAQ membership). In both panels, the sample period covers M1:1989–M12:2010.* indicates significance at the 10% level; ** indicates significance at the 5% level; and *** indicates significance at the 1% level.

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Korniotis and Kumar 2013). We thus run two additional models, in which we condition on states with above median local bias in the trading decisions of either both institutional and retail investors or retail investors only. State-level estimates of local bias are taken from Korniotis and Kumar (2013). Moreover, we run a tone-enhanced analysis similar to Table 8. The major difference is that we now rely on local (as opposed to national) media and also orthogonalize the tone with respect to state dummies to account for potential cross-sectional differences in the average tone (see Table 1).

Panel B shows a pervasive picture: the interaction term between residual media coverage and the individualism dummy is always positive and often statistically significant, suggesting a stronger impact of the media on return predictability in individualistic states. Moreover, sharpening the analysis with respect to cross-sectional differences in local bias or article tone leads to stronger estimates. Finally, findings are also meaningful from an economic perspective. For instance, coefficients suggest that the impact of a one-standard-deviation change of tone-enhanced residual media coverage on media-based momentum is, all else equal, about 10 to 15 bp per month stronger in individualistic regions than in collectivistic regions.

Chui, Titman, and Wei (2010) interpret their cross-country evidence as support for Daniel, Hirshleifer, and Subrahmanyam's (1998) framework, and it appears natural to do the same for our within-country results. Further support for the idea that the Vandello and Cohen (1999) index proxies for biases important in the Daniel, Hirshleifer, and Subrahmanyam (1998) framework comes from recent corporate finance studies that find the index to be correlated with measures of managerial overconfidence (Chen et al. 2012; Truong et al. 2013).

4. Conclusion

Although the momentum effect is among the most prominent return anomalies, its underlying drivers are far from being well understood. We provide some novel evidence to the ongoing debate by exploiting a unique data set comprising about 2.2 million firm-specific articles from forty-five newspapers.

Our findings reveal a systematic link between the extent of a firm's excess media coverage and the magnitude of the momentum and long-term reversal effect in its stock. Consistent with the idea that investor overconfidence should strengthen the effect, we document higher media-based momentum profits among hard-to-value firms, among firms for which the article tone matches the formation period return, and among firms located in more individualistic U.S. states.

At least with respect to our relatively large sample firms, these insights lend support to the idea that the momentum effect may, to a substantial extent, be an overreaction-driven phenomenon. In this way, return predictability can be strongest for those firms that stand particularly in the public spotlight. In

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a broader sense, our findings suggest that a joint analysis of cross-sectional differences in information transmission and investor biases may help to develop a better understanding of return anomalies.

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